Use of Headphones and its Impact on Hearing: A Cross-Sectional Survey of Medical, Nursing, and Pharmacy Students

Gursimrat Kaur, Aditya Anand, Priyanko Chakraborty

1. Otolaryngology - Head and Neck Surgery, Mata Gujiri Memorial National College and LSK Hospital, Kishanganj, IND

Corresponding author (Research Outputs): Priyanko Chakraborty, cpriyanko@gmail.com

ABSTRACT

Background

Personal audio device use among healthcare students is widespread, raising concerns about noise-induced hearing loss.

Objective

To evaluate patterns of headphone use, quantify noise exposure, and examine their association with auditory symptoms and sensorineural hearing loss (SNHL) among medical students pursuing the Doctorate of Medicine, MBBS degree, pharmacy, and nursing students

Methodology

A cross-sectional survey surveyed 250 undergraduate and graduate medical, pharmacy, and nursing students (aged 18-30, mean 20.8 +/- 2.1) at MGM Medical College and Hospital, Kishanganj. Participants completed a questionnaire on headphone type, listening duration, and volume, followed by otoscopic evaluation and pure-tone audiometry. Associations between device-use patterns and auditory outcomes were tested using the chi-square test.

Results

Regular headphone use was reported by 210,X students (84.0%). Earbuds were the most common device (91 users, 43.3%), followed by wired earphones (58, 27.6%), and headsets (14, 6.7%). Daily listening lasted < 2 hours for 92 users (43.8%), 2-4 hours for 85 (40.5%), 4-6 hours for 25 (11.9%), and > 6 hours for 8 (3.8%). Volume was kept below 50% by maximum of 100 users (47.6%), at 50-80% by 103 (49.0%), and above 80% by 7 (3.4%).

Conclusions

Unsafe listening practices are common among healthcare students. Higher listening volumes and extended daily use were associated with increased auditory symptoms and early SNHL. Volume regulation appears more critical than duration alone in reducing risk. Educational

initiatives and routine screening are warranted, particularly for MBBS students, to protect hearing health.

Categories: Health Policy, Medical Education, Otolaryngology

Introduction

In recent years, the use of personal audio devices such as headphones and earphones has become increasingly prevalent among students, particularly in demanding academic fields related to health care. These devices are not only used for entertainment but also for educational purposes, including listening to lectures, participating in discussions, and attending virtual listening sessions. While convenient, this growing dependence on headphones raises serious concerns regarding noise exposure and its potential impact on hearing health [1].

Many commercially available headphones and earphones have the potential to generate sound levels exceeding 125 decibels (dB)[2], far surpassing the recommended safe learning threshold [3]. Extended use of these devices at elevated volume levels can lead to cumulative trauma to the delicate cochlear hair cells, ultimately resulting in sensorineural hearing loss (SNHL) - particularly affecting the perception of high-frequency sounds. Such damage is permanent and may go permanently unnoticed in early stages, making it a silent but serious health concern [4]. Globally, the prevalence of disabling hearing loss among adults is around 16%, with regional variations ranging from 7% to as high as 21%, depending on demographic and environmental factors [5,6]. Despite these concerning statistics, hearing loss when especially when caused by recreational noise exposure, remains underrecognized and underreported, particularly in younger populations.

College students may not be fully aware of the long-term consequences of unsafe listening practices. Continuous exposure to high-volume sound through headphones or earphones, listening devices, combined with a lack of knowledge about safe usage guidelines, makes this population particularly vulnerable to audio damage.

This study aims to explore the patterns of headphone use, assess the extent of noise exposure, and evaluate the level of hearing-risks and problems among medical, pharmacy, and nursing students. With the increasing use of personal audio devices in academic and recreational settings, this study investigates how listening habits may be contributing to early signs of hearing impairment. By addressing this often-overlooked aspect of student well-being, this research means to provide meaningful some insights and promote preventive strategies for hearing conservatism.

Materials and Methods

A cross-sectional study was conducted to evaluate headphone use and noise exposure among graduate students enrolled in the Doctorate of Medicine, Doctorate of Surgery (MBBS), Pharmacy, and Nursing Programs. The study spanned six (6) months in the Outpatient Department (OPD) of MGM Medical College and Hospital, Kishanganj. Eligible participants 18-30 were whom provided written informed consent in WI, CA, and KY only. Exclusion criteria included pre-existing hearing loss, hearing aid usage, prior ear surgery, or inability to complete the 1-year or 2-year study process.

Sample Size Calculation

Because the target population was students, the required sample size was calculated using a pooled prevalence of hearing loss among college students (19%) reported by a recent meta-analysis [8]. The single-proportion formula was applied: n = (Z^2 * p(1-p))/d^2, with Z = 1.96 (95% confidence), p = 0.19, and d = 0.05. This yielded n = 236.5, rounded to 236. Allowing for non-response, 250 students were enrolled.

Data collection

The structured questionnaire was developed by the authors after a literature review and pilot-tested among 20, 251,250,251,20 students for internal consistency before full-scale administration. It captured demographics, headphone usage (duration, frequency, type, and volume), and noise exposure history (Appendix). Clinical otoscopic evaluation was performed to exclude external or middle ear pathology.

Audiometric assessment

All participants underwent pure tone audiometry (PTA) in a sound-tested room using a calibrated ALPS AD1200 clinical audiometer (ALPS, New Dehli, India; ANSI 53.6-2018 standard), with TDH-39 (AT-541,AT-540,AT-543,AT-545 GM) TDH-39 supraural headphones (Telephonics Corporation, *do not interfere with Raghav Rathi*, Farmingdale, NY). Tests were performed by trained audiologists approximately 24 hours after the last headphone use to minimize temporary threshold shifts. Hearing thresholds were assessed from 250 to 8,000 Hz in both ears, and results interpreted according to standard criteria to identify hearing loss.

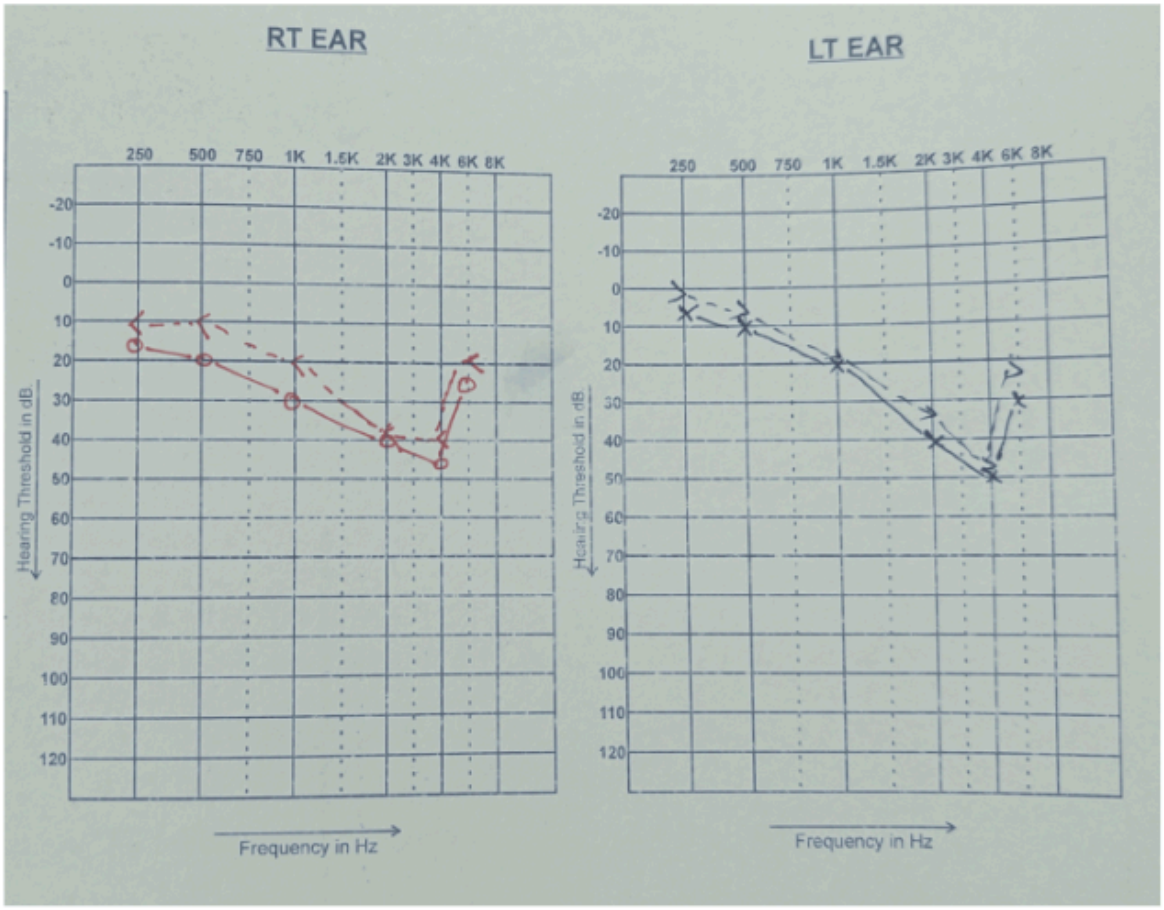

FIGURE 5: Audiogram demonstrating a 4-6kHz notching pattern typical of noise-induced hearing loss.

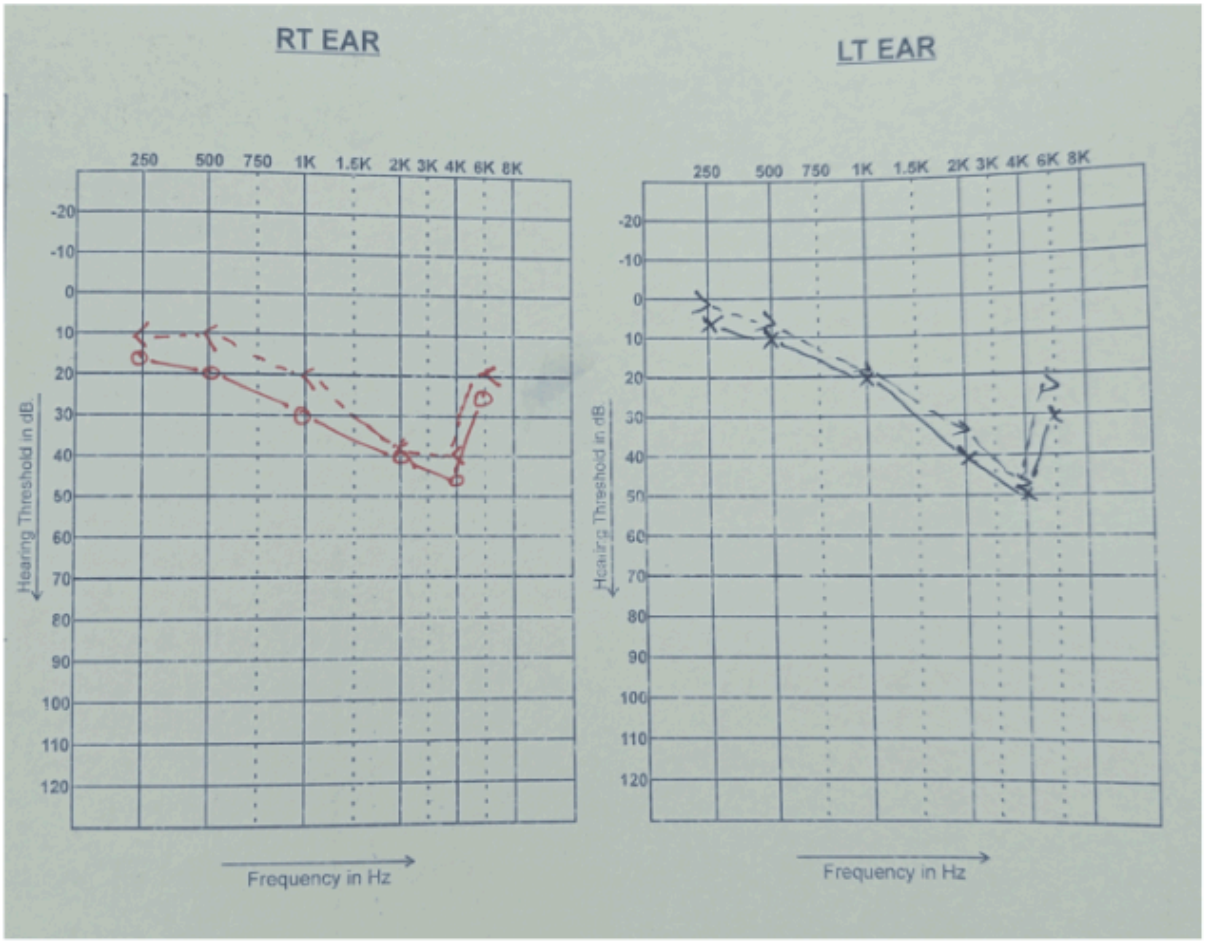

FIGURE 5: Audiogram demonstrating a 4-6 kHz notching pattern typical of noise-induced hearing loss.

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Results

A total of 250 students participated in the study, comprising 121 (48.4%) Medical (MBBS), 56 (22.4%) Pharmacy, and 73 (29.2%) Nursing students. The mean age was 20.8 ± 2.1 years, with a range of 18-30 years. Table *1* presents the demographic profile across academic programs. Participants were primarily enrolled in Medical (MBBS), Pharmacy, and Nursing programs, with Medical (MBBS) students constituting the largest subgroup. Gender distribution differed significantly among courses ($\chi^2 = 92.2$, $P < 0.001$).

Academic program	Number of students (*n*)	Percentage (%)	Gender distribution (M/F)	Average age (Years)	Test statistic (χ^2)	*P*-value
Medical	121	48.4	74/47	20.9 ± 1.8	$\chi^2 = 92.2$	<0.001
Pharmacy	56	22.4	42/14	20.2 ± 2.3		
Nursing	73	29.2	0/73	21.5 ± 1.6		

TABLE 1: Distribution of students by program, gender, and age (n = 250).

The chi-square test was used to assess differences in gender distribution across academic programs.

Out of the 250 participants, 210 students (84%) reported regular use of headphones. Among these 210 students, earbuds were the most common device (91, 43.3%), followed by wired earphones (58, 27.6%), Bluetooth earphones (47, 22.4%) and headsets (14, 6.7%). The distribution of device preferences varied across academic programs (Table *2*, Figure *1*). Medical (MBBS) students predominantly used earbuds (57, 47.1%), whereas nursing students showed a marked preference for wired earphones (35, 47.9%). Pharmacy students demonstrated a more balanced distribution across device types, though earbuds remained the most popular (19, 33.9%).

Course	Earbuds, *n* (%)	Bluetooth, *n* (%)	Wired, *n* (%)	Headset, *n* (%)	None, *n* (%)	Test statistic (χ^2)	*P*-value
Medical	57 (47.1)	25 (20.7)	13 (10.7)	5 (4.1)	21 (17.4)	$\chi^2 = 40.88$	<0.001
Pharmacy	19 (33.9)	12 (21.4)	10 (17.9)	5 (8.9)	10 (17.9)		
Nursing	15 (20.5)	10 (13.7)	35 (47.9)	4 (5.5)	9 (12.3)		

TABLE 2: Distribution of device types used among healthcare students by course (n, %).

Q. No.	Question	Options (tick ☐ one)
1	**Age: _____ Sex: ☐ Male ☐ Female**	
2	Course & Year: _____	
3	Daily mobile/iPad screen time	☐ < 4 h ☐ 4–6 h ☐ 6–8 h ☐ > 8 h
4	Use of Headphones/Earbuds/Bluetooth	☐ Yes ☐ No
5	Daily headphone use duration	☐ < 2 h ☐ 2–4 h ☐ 4–6 h ☐ > 6 h
6	Usual volume level	☐ < 50% ☐ 50–80% ☐ > 80%
7	Do you play mobile games?	☐ Yes ☐ No
7a	If Yes, gaming duration	☐ < 4 h ☐ 4–6 h ☐ 6–8 h ☐ > 8 h
8	Game played most often	_____
9	Symptoms experienced (tick all that apply)	☐ Tinnitus ☐ Hearing difficulty ☐ Loud sound intolerance ☐ Headache ☐ Ear pain ☐ Eye watering ☐ Insomnia
10	Any ear problem (past or present)	☐ Yes (specify) _____ ☐ No
11	History of ear surgery	☐ Yes ☐ No
12	Usual bedtime	_____
13	Sleep duration in 24 h	_____ hours
14	Willing for free hearing assessment (PTA / Impedance audiometry)?	☐ Yes ☐ No

TABLE 5: Performa (Survey Tool / Data Collection Sheet)